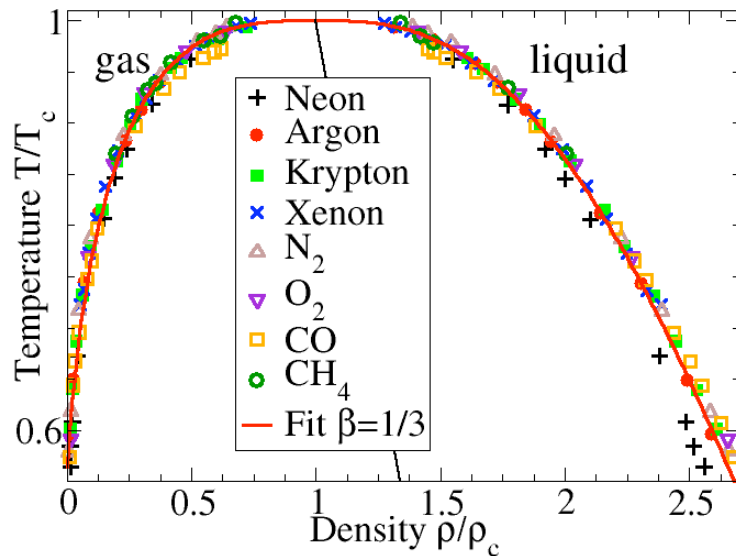
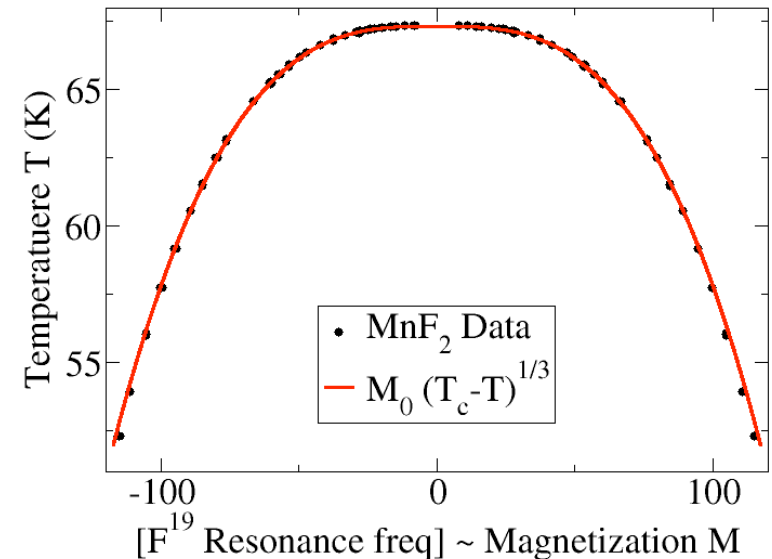


Universality: Shared Critical Behavior

Ising Model and Liquid-Gas Critical Point



Same critical
exponent
 $\beta=0.332!$



Liquid-Gas Critical Point

$$\rho - \rho_c \sim (T_c - T)^\beta$$

$$\rho^{Ar}(T) = A \rho^{CO}(BT)$$

Ising Critical Point

$$M(T) \sim (T_c - T)^\beta$$

$$\rho^{Ar}(T) = A(M(BT), T)$$

Universality: Same Behavior up to Change in Coordinates

$$A(M, T) = a_1 M + a_2 + a_3 T + \text{(other singular terms)}$$

Nonanalytic behavior at critical point (not parabolic top)

All power-law singularities (χ , c_v , ξ) are shared by magnets, liquid/gas